

THEME

THEMES (ORDERED BASED ON PRIORITY AS PER OUR MEETING WITH UTRENTO)

Theme	Description	Relevance	AI & Citizen Science
Youth Employment, Skills & the Future of Work	Youth map employment realities, skill gaps, and entrepreneurship opportunities.	High youth unemployment and informal work dominate the region.	Surveys and AI-based analysis help identify trends and inform solutions.
Digital Inclusion, AI Literacy & Ethics	Digital Inclusion, AI Literacy & Ethics	Digital divides remain strong, risking exclusion from the digital economy.	Participants test AI tools and reflect on responsible use.
Gender Equality & Social Inclusion	Youth document barriers faced by women and marginalized groups.	Gender inequality and social exclusion remain critical challenges.	What youth collect - Public spaces where women and men participate differently (parks, universities, workplaces) - Access points to services (health centres, transport, community centres) - Visibility of women in leadership or public roles (posters, signage, events) - Barriers to inclusion (inaccessible buildings, unsafe streets – <i>without people's faces</i>) - Narratives on women on social media/newspapers/ radio, etc Locations perceived as: - Safe/unsafe for women - Inclusive/exclusive - Accessible/inaccessible

Sustainable Cities & Local Innovation	Youth explore urban challenges such as housing, mobility, and energy.	Rapid urbanisation affects the quality of life.	Mapping and data visualisation support local innovation.
Public Health, Wellbeing & Mental Health	Youth investigate access to healthcare and mental health challenges.	Mental health is under-researched and stigmatized.	Analysing anonymous surveys with AI reveals patterns.
Civic Engagement, Trust & Democratic Participation	Youth study participation in civic life, trust in institutions, and misinformation.	Youth voices are often excluded from decision-making.	Data collection and sentiment analysis support civic understanding.
Climate Change, Environment & Local Resilience	Youth explore local climate impacts such as flooding, droughts, waste management, and coastal erosion.	Sub-Saharan Africa is highly vulnerable to climate change. These issues directly affect daily life in Nigeria, South Africa, and Liberia.	Youth collect geo-tagged environmental data and use AI tools to analyse trends and risks.

TOPICS

Level	Theme	Project topic	Student's name	Supervisor/Co-supervisor
Masters		IsiZulu Generative AI with Emerging Technology Vocabularies Application	Kwanele Mazibuko	Linde Manqele/Prof Ojo
Masters		IsiZulu Generative AI for Crop and Pest Detection		Freedom Khubisa/Prof Olugbara
PhD		Spatio-Temporal Crime Hotspot Identification and Prediction Model	Shumi Chabalala	Profs Ojo, Olugbara & Giunchiglia
		Behavioral insights and hotspot identification: Integrating NLP, ML, and Geospatial analysis of automobile crashes		
		IsiZulu GenAI for Carbon Footprint hotspots detection in Computer Labs		
		IsiZulu GenAI for XCO2 hotspots detection in urban/Industrial areas		
		LLM-enhanced argument extraction and clustering model for spatio-temporal urban hotspot event detection using crowdsourced data		
		More to be added		